coalesce after a time $(5/512) \times (c^5 / G3)(D^4/M_h^3)$. G is Newton's gravitation constant and c is the speed of light; see above. See, e.g., Equation (36.17b) of MTW.

53 [The ring has a circumference of 5 million kilometers, . . . curvature of spacetime.] A person on the girder-work ring at a distance L from its central layer feels an acceleration $a = (24\pi^3 GM_h/C^3)L$ toward the central layer, caused one third by the rotating ring's centrifugal force and two thirds by the hole's tidal force. G is Newton's gravitation constant, M_h is the hole's mass, and C is the circumference of the ring's central layer. For comparison, 1 Earth gravity of acceleration is 9.81 meters per second^2. See the note for page 35 above.

55 [The laws of quantum gravity . . . usable for time travel.] 1.62×10^{-33} centimeter = $= \sqrt{G\hbar/c^3}$ is the "Planck-Wheeler length," with G = Newton's gravitation constant, c = the speed of light, and $\hbar$ = Planck's constant (1.055×10^{-34} kilogram-meter^2 per second). See pages 494 and 518 of Chapter 14.

57–58 [Another is the fact that, . . . flying colors.] See, e.g., Will (1986).

CHAPTER 1

59 General comment about Chapter 1: Most of this chapter's material about Einstein's life comes from the standard biographies of him: Pais (1982), Hoffman (1972), Clark (1971), Einstein (1949), and Frank (1947). For most of the historical perspective and quotations in Chapter 1, which I have gleaned from these standard biographies, I do not give individual citations below. Much new historical material is becoming available with the gradual publication of Einstein's collected papers, ECP-l, ECP-2, and Einstein and Maric (1992). I do cite, below, material from these sources.

59–60 [Professor Wilhelm Ostwald . . . Hermann Einstein.] Document 99 in ECP-1.

60 ["Unthinking respect . . . enemy of truth,"] Document 115 of ECP-l, as translated on page xix of Renn and Schulmann (1992).

61 Footnote 1: The following example illustrates what is meant by "mathematically manipulating" the laws of physics.

Early in the seventeenth century, Johannes Kepler deduced, from Tycho Brahe's observations of the planets, that the cube of the circumference C of a planet's orbit divided by the square of its orbital period P, i.e., C^3/P^2, was the same for all the planets then known: Mercury, Venus, Earth, Mars, Jupiter, Saturn. A half century later, Isaac Newton explained Kepler's discovery by a mathematical manipulation of the Newtonian laws of motion and gravity (the laws listed on page 61 of the text):

1. From the following diagram and a fair amount of sweat, one deduces that, as a planet encircles the Sun, the planet's velocity changes at a rate given by the formula, (rate of change of velocity) = $2\pi C/P^2$, where $\pi = 3.14159...$. This rate of change of velocity is sometimes called the *centrifugal acceleration* that the orbiting planet experiences.